

■ Phase 2: Requirement Analysis

Technology Stack

Date:	July 3, 2026
Team ID:	SWTID-2026-4823
Project Name:	BOOK STORE (BookNest)

Technical Architecture

The platform is built using a MERN (MongoDB, Express, React, Node) stack. The design implements a 3-tier model separating the Client Interface, Backend REST API, and Database Layer. Communication is handled over HTTPS via JSON payloads.

Table-1: Components & Technologies

S.No	Component	Description	Technology Stack
1	User Interface	Single Page Application rendering dashboards for user, seller, and admin roles.	React.js (Vite), HTML5, CSS3, React Router DOM
2	Application Logic	RESTful API processing core operations, routing and validation middleware.	Node.js, Express.js
3	Authentication	Secure password hashing and stateless token-based session management.	JSON Web Tokens (JWT), bcryptjs
4	File Uploads	Handles book cover image multipart uploads, storing them in public folders.	Multer middleware, Local server filesystem
5	Database	Document-based relational data modeling and database queries.	MongoDB, Mongoose ODM
6	Cloud Hosting	Cloud platforms hosting frontend files, REST API, and data storage.	Vercel (Client), Render (Backend), MongoDB Atlas (Cloud Database)

Table-2: Application Characteristics

S.No	Characteristic	Description	Technology Used
1	Open-Source	The system uses free, modular, and modern open-source node packages.	React, Express, Mongoose, Vite
2	Security	Encrypt credentials, enforce auth middleware, configure CORS policies.	bcrypt (salt=10), JWT verification, CORS headers
3	Scalability	Stateless REST API allows horizontal scaling across backend nodes.	RESTful architecture, stateless JSON payloads
4	Availability	Multi-region replica sets prevent data loss during single node failures.	MongoDB Atlas Cluster replication
5	Performance	Quick rendering, caching, and minimum query load.	Vite bundling, Mongoose indexing on book fields (title, author, tags)